

Build a Fair Probability Model



Grade 7 • Mathematics
NC.7.SP.7

Use equally likely outcomes to model chance.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A bag holds 3 red, 4 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

There are 10 equally likely counters. Red: $3/10$; not red: $7/10$. These disjoint events cover all 10 counters.



Build a second model together

A bag holds 3 red, 7 blue, and 5 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Discuss: What should the first step find?



Your turn • 1

A bag holds 8 red, 3 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Represent → solve → explain → check



Your turn • 2

A bag holds 3 red, 3 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Represent → solve → explain → check



Your turn • 3

A bag holds 2 red, 8 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Represent → solve → explain → check



Your turn • 4

A bag holds 3 red, 8 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Represent → solve → explain → check



Exit question • Explain independently

A bag holds 7 red, 3 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 4/9; 5/9

Exit: 7/12; 5/12

2. 3/13; 10/13

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

3. 2/17; 15/17

4. 3/14; 11/14

